

AFFAN MUHAMMAD SHAIKH

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EDUCATION

Ontario Tech University

Oshawa, ON

Bachelor of Information Technology (Honours) - Networking and IT Security

Sept. 2024 - Expected Apr. 2028

- Relevant coursework: Advanced Networking I, Cybersecurity Foundations, Programming I with Python, and Computer Systems.

PROJECTS

Cisco Networking Labs | *Cisco IOS, Packet Tracer, Wireshark*

- Built and verified routed and switched Cisco topologies using IPv4/IPv6 addressing, VLANs, access ports, 802.1Q trunks, inter-VLAN routing, DHCP, STP, NAT, and OSPF/EIGRP fundamentals.
- Diagnosed connectivity and routing faults with IOS show commands, interface state, routing tables, ping, traceroute, and packet captures; validated end-to-end reachability after each configuration change.

P2P Messaging (Collaborative WIP) | *Python, FastAPI, Ed25519, X25519, TCP*

[GitHub](#)

- Co-developed a direct encrypted-messaging prototype with password-protected Ed25519/X25519 identities, out-of-band fingerprint verification, ChaCha20-Poly1305 encryption, signed envelopes, and authenticated acknowledgements.
- Built a local browser workspace with device unlock, separate conversations, authenticated replies, history search, CSRF-protected actions, encrypted local history, and attachment digest binding; enforced recipient and expiry checks, 64 KiB frame and 4 KiB plaintext limits, and persistent replay rejection.

Secure File Transfer | *Python, TLS, SHA-256, pytest*

[Case Study](#)

- Built an authenticated TLS file-transfer service and local browser workspace for certificate setup, account management, server control, file selection, recipient inboxes, and verified downloads without storing UI passwords.
- Implemented recipient-scoped storage, resumable transfers, SHA-256 re-hashing, quarantine, script password records, throttling, and audit logs; verified eight upload/download round trips totalling 13,632,512 bytes and resumed a 2 MiB upload from the exact 700,000-byte server offset.

File Integrity Monitor | *Python, SHA-256, JSON, unittest*

[GitHub](#)

- Built a dependency-free integrity application with a loopback-only browser dashboard and CLI automation for trusted SHA-256 baselines, grouped change review, deterministic JSON evidence, exclusions, and script-friendly exit codes.
- Detected 45 of 45 controlled changes across 500 fixture files with zero scan errors, including 20 modifications, 10 deletions, 10 additions, and five moves; also handled same-size tampering, symbolic-link safety, and deterministic evidence persistence.

OTNow Canvas Companion | *Chrome Manifest V3, JavaScript, Canvas APIs*

[GitHub](#)

- Built a local-first Ontario Tech Canvas extension with grouped deadlines, moved-date detection, reminders, course links, offline cache, and dark mode; submitted v1.5.0 for Chrome Web Store review and built an opt-in public dashboard currently reporting 200+ privacy-rounded Canvas items organized.

TECHNICAL SKILLS

Certification: CompTIA Security+ in progress

Security: TLS, ChaCha20-Poly1305, Ed25519/X25519, SHA-256, scrypt password records, file-integrity monitoring, access control, audit logging, RBAC, SSRF defenses

Networking: IPv4/IPv6, subnetting, VLANs, 802.1Q, inter-VLAN routing, DHCP, DNS, NAT, STP, OSPF/EIGRP fundamentals, routing and switching troubleshooting

Programming: Python, FastAPI, pytest, unittest, JSON, TypeScript, JavaScript

Systems and Tools: Linux, Windows Server, Cisco IOS, Wireshark, Packet Tracer, Git, GitHub, Cloudflare Pages

EXPERIENCE AND LEADERSHIP

Co-Founder and Platform Developer

May 2026 - Present

SSIK IT Consulting & Solutions

Ontario, Canada

- Co-founded an IT and cybersecurity consultancy with Ontario Tech classmate Ghayas Sher; jointly develop service scopes, security-control review methods, privacy research, and stakeholder-ready recommendations.
- Built the public website and private SSIK Intelligence platform; delivered a 12-stage local workflow with multi-workspace RBAC, SSRF-hardened passive collection, durable jobs, evidence-linked review, approval-gated mock outreach, rescans, backup, and recovery controls.

Technical Operations and Hosting

2026 - Present

Archtech - Developing Nonprofit

Oshawa, ON

- Set up Google Workspace and manage website hosting and deployment while supporting implementation for an unreleased nonprofit platform.
- Coordinate contributors through a private shared repository and communicate hosting, access, and deployment requirements as the team works toward launch.